

# ATHER SAYED

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

Machine Learning Engineer with end-to-end experience in Python, TensorFlow, PyTorch, FastAPI, and Docker, specializing in deep learning, computer vision, NLP, and MLOps. B.Sc. Electrical Engineering (2024 Graduate) who dedicated the post-graduate year to an intensive technical specialization, earning 13 industry certifications from Stanford, DeepLearning.AI, and IBM, including Generative AI and LLMs for NLP. Proven expertise in building end-to-end ML pipelines, transfer learning for medical imaging, handling imbalanced datasets, and shipping containerized REST APIs — highlighted by a full-stack Credit Card Fraud Detection system. Strong focus on deploying scalable, high-accuracy models in production.

## TECHNICAL SKILLS

**Machine Learning:** Supervised & Unsupervised Learning, Deep Learning, CNNs, Transfer Learning, Reinforcement Learning, XGBoost, Anomaly Detection, Model Evaluation, Hyperparameter Tuning

**Generative AI & NLP:** Large Language Models (LLMs), Generative AI, Transformer Architecture, Foundational Models, Natural Language Processing

**Computer Vision:** Image Classification, Medical Imaging, OpenCV, Data Augmentation, ONNX Inference

**MLOps & Backend:** FastAPI, REST APIs, Docker, Docker Compose, PostgreSQL, SQL, Git, GitHub, Linux, Model Deployment

**Languages & Libraries:** Python, TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Hugging Face

**Foundations:** Linear Algebra, Calculus, Probability & Statistics, Data Preprocessing, Feature Engineering

## EDUCATION

**Bachelor of Science in Electrical Engineering — Computer and Systems** | 09/2020 – 09/2024

Ain Shams University, Cairo, Egypt | Very Good

## PROJECTS & EXPERIENCE

**Full-Stack Credit Card Fraud Detection System — Python, FastAPI, PostgreSQL, Docker, React** | [GitHub](#)

- Designed and shipped an end-to-end fraud detection platform pairing an unsupervised Isolation Forest model with a real-time scoring API and an analyst-facing dashboard.
- Built a FastAPI backend that scores transactions on demand and persists every flagged event to a PostgreSQL Incident Log for auditing and review.
- Developed a React + Vite + Tailwind + shadcn/ui dashboard for live monitoring, transaction inspection, and incident triage.
- Containerized the full stack (DB, API, UI) with Docker Compose for one-command, reproducible deployment.
- Documented data exploration, feature engineering, and model training in Jupyter notebooks.

**Breast Cancer Detection using Deep Learning — Python, TensorFlow/Keras, OpenCV, ONNX** | [GitHub](#)

- Built a transfer learning pipeline (VGG16) to detect Invasive Ductal Carcinoma in histopathology images with two-phase fine-tuning.
- Applied OpenCV for image preprocessing and ImageDataGenerator for data augmentation on 280,000+ image patches.
- Exported the final model as ONNX for production-ready, framework-agnostic inference.
- Result: Achieved ~94% accuracy, 0.99 AUC, and ~98% recall — critical for minimizing missed cancer diagnoses.

**House Prices Prediction — Python, XGBoost, Scikit-learn, Pandas** | [GitHub](#)

- Developed a regression model using advanced feature engineering and ensemble methods (XGBoost, Random Forest) on structured tabular data.
- Performed exploratory data analysis, handled missing values, and applied log transformations to improve model accuracy.

**Digit Recognizer with CNN — Python, TensorFlow/Keras** | [GitHub](#)

- Implemented a Convolutional Neural Network to classify handwritten digits from the MNIST dataset with high accuracy using TensorFlow/Keras.

## CERTIFICATIONS

- Supervised ML: Regression and Classification | DeepLearning.AI & Stanford | 12/2025
- Advanced Learning Algorithms | DeepLearning.AI & Stanford | 01/2026
- Unsupervised Learning, Recommenders, Reinforcement Learning | DeepLearning.AI & Stanford | 01/2026
- Neural Networks and Deep Learning | DeepLearning.AI | 02/2026
- Machine Learning with Python | IBM | 03/2026
- Intro to Deep Learning & Neural Networks with Keras | IBM | 03/2026
- Deep Learning with Keras and TensorFlow | IBM | 04/2026
- Introduction to Neural Networks and PyTorch | IBM | 04/2026
- Deep Learning with PyTorch | IBM | 04/2026
- AI Capstone Project with Deep Learning | IBM | 05/2026
- Generative AI and LLMs: Architecture and Data Preparation | IBM | 05/2026
- Gen AI Foundational Models for NLP & Language Understanding | IBM | 06/2026
- Docker Training Course for the Absolute Beginner | KodeKloud | 04/2026

## LANGUAGES

Arabic (Native)   English (Professional Proficiency)